

Assessment

Module 8: JavaScript Essentials And Advanced

Case Overview

Scenario:

You are an intermediate JavaScript developer at **Hashcrypt technologies Pvt. Ltd.** The company wants a smart, error-resilient, and modular **Task Manager Web App**. The app must be responsive, interactive, and follow good programming practices.

Core Functionality	
1. Responsive Form Creation	○ Task creation form with fields: ▪ Task Title (text) ▪ Description (textarea) ▪ Due Date (date) ▪ Priority (select: High, Medium, Low) ○ Status (select: To Do, In Progress, Done) ○ Form validation (required fields) ○ Task display in a list ○ Edit and Delete task actions ○ Persist tasks using localStorage ○ Real-time UI update on actions ● Mobile-friendly layout using Flexbox/Grid and media queries
2. Form Handling with Advanced JavaScript	Class: CustomerFormHandler ○ validateForm() – Validate: ▪ task title ≥ 10 characters ▪ Descriptions ▪ Due date

	▪ Priority ▪ status ○ saveToLocalStorage() – Save data as array in localStorage ○ clearForm() – Reset form after submission showMessage() – Reusable UI message box for success/error Additional Requirements: ● Use event delegation for handling form events ● Implement real-time validation (on input or blur)
3. View Submissions Page (view.html)	Class: SubmissionViewer ● Read from localStorage ● Dynamically generate table of submissions ● Search bar to filter by Name or Vehicle ● Show “No data found” message if results empty

Optional Bonus Tasks

- Add **Delete** button to remove individual records
- Use **utility module** for setItem() and getItem() (modularized code)
- Implement **import/export** (<script type="module"> or bundler like Vite/Webpack)

Validation Rules

- All fields required
 - Descriptions
 - Due date
 - Priority
 - status
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Key Competencies Tested

- ES6 Classes & Modules
 - DOM Manipulation & Event Handling
 - Real-Time & Manual Form Validation
 - LocalStorage CRUD Operations
 - Dynamic HTML Rendering & Filtering
 - Clean, Modular, and Reusable Code
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Reflective Thinking Questions

- How could this app be enhanced by integrating with an **API** (e.g., saving data to a backend)?
- How can error handling be improved for a better **user experience**?
- What security concerns arise when storing customer data in localStorage?
- Could the app be scaled into a **PWA** (Progressive Web App) for offline use?